User Instructions/Guide

Topic: DrillMain Intelligence

Smart drilling insights with next-step guidance and integrated maintenance.

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Course: CSIS 4495 002 Applied Research Project

Section: 002

Team Lead: Uchechi Sebastian (Individual Project)

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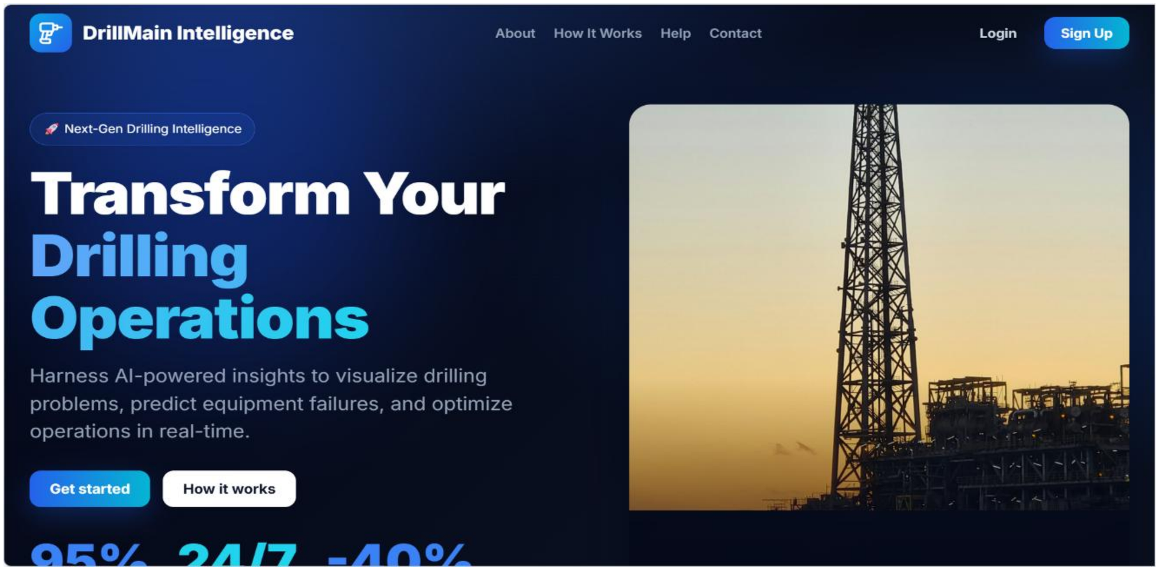
Home Page

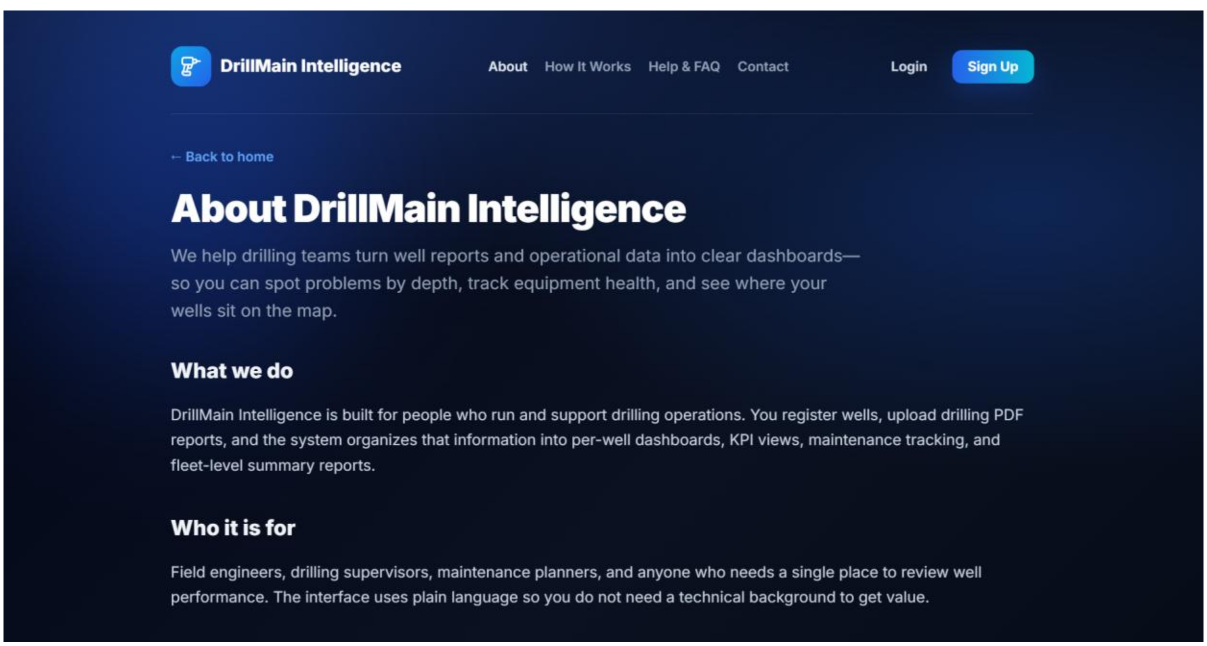
1.1 Home screen:

- The first thing you see on your screen after loading the website is the home page (see pictures below).



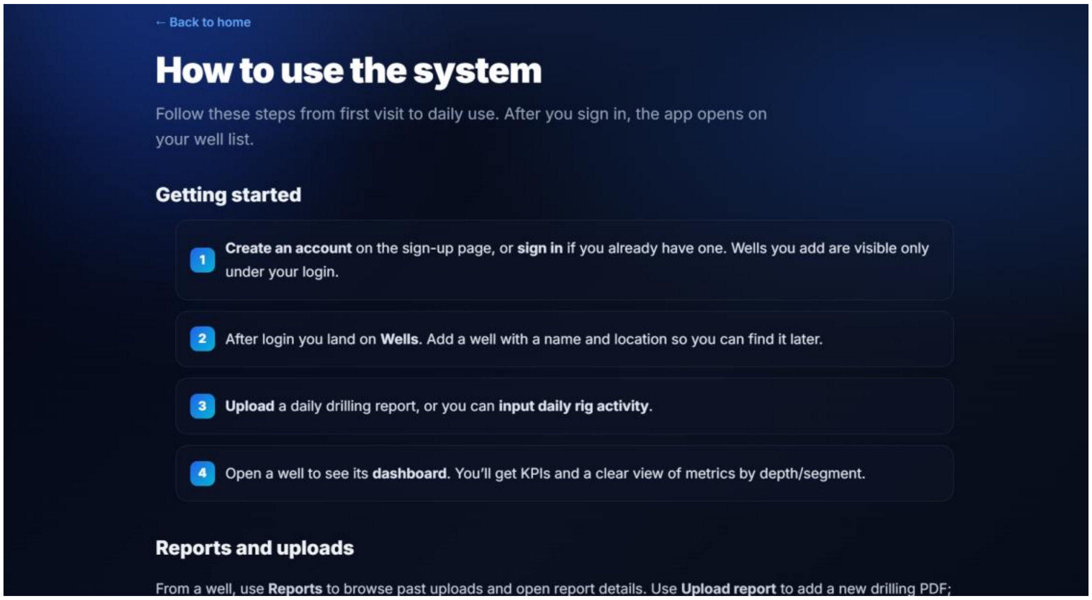
1.2 About Page:

- On the Home page, click on “About” link at the top of the page. This page explains what DrillMain Intelligence gives you a brief introduction of what the app is all about.



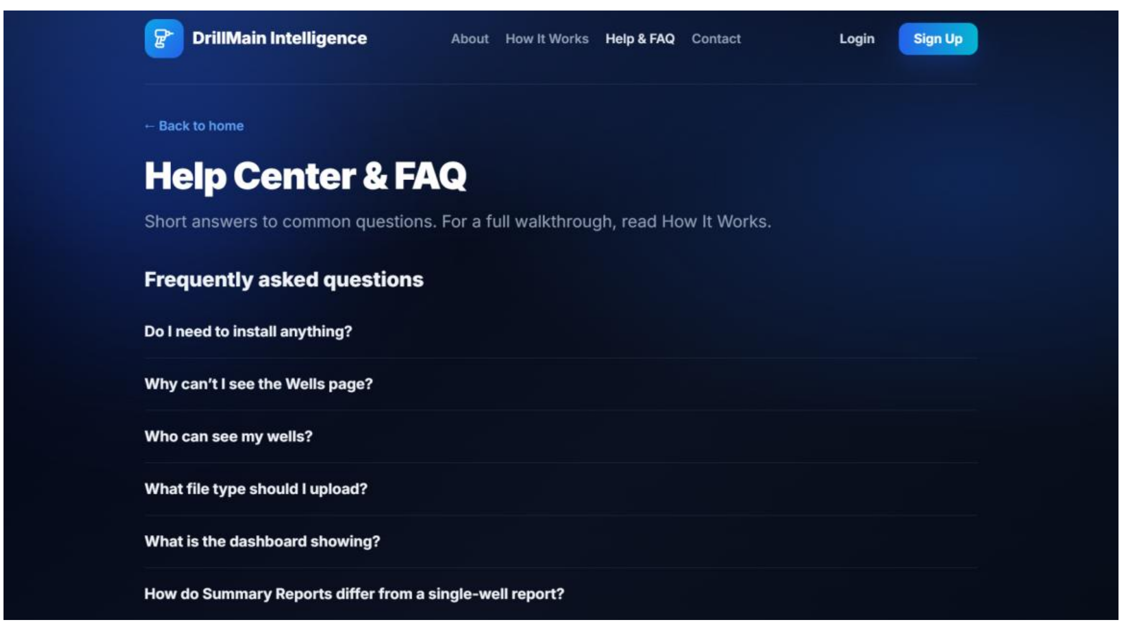
1.3 How it works:

- On the Home page, click on “How it works” link at the top of the page. This page gives a brief overview on how to use the system.



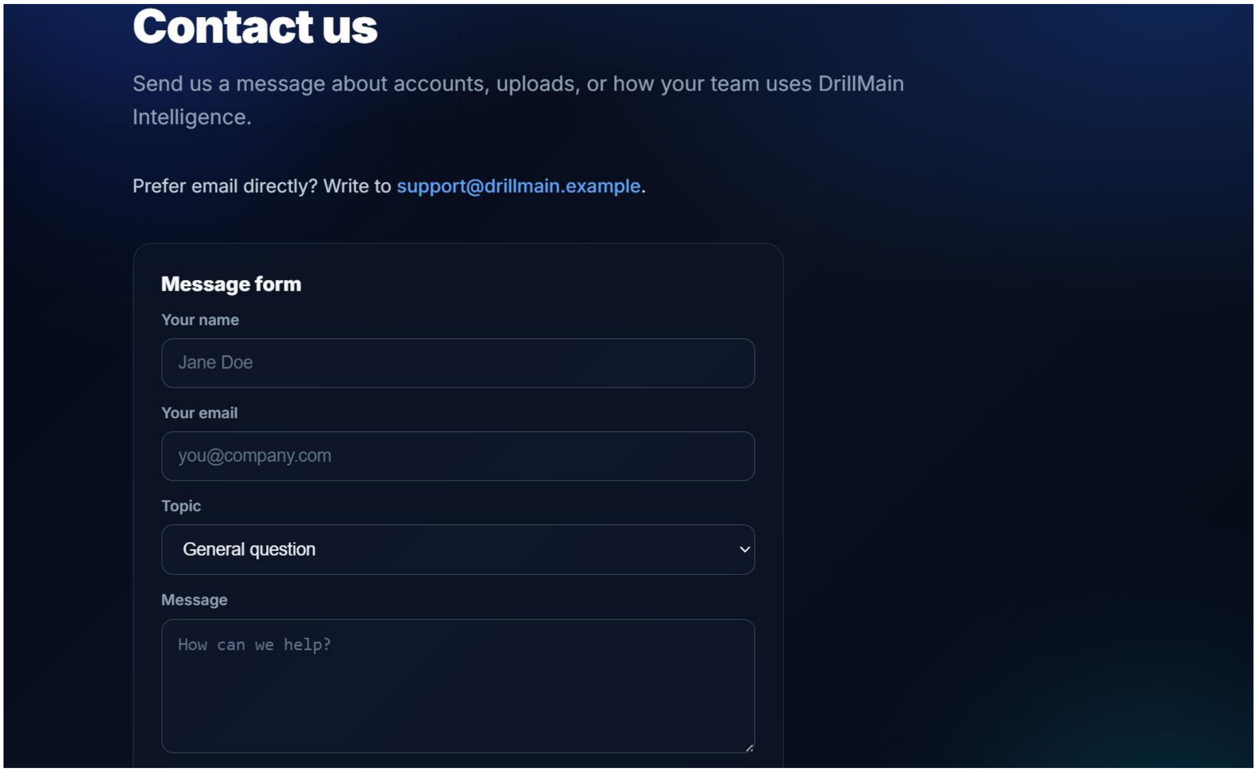
1.4 Help & FAQ:

- On the Home page, click on “About” link at the top of the page. This page explains what DrillMain Intelligence gives you a brief introduction of what the app is all about.



1.5 Contact us:

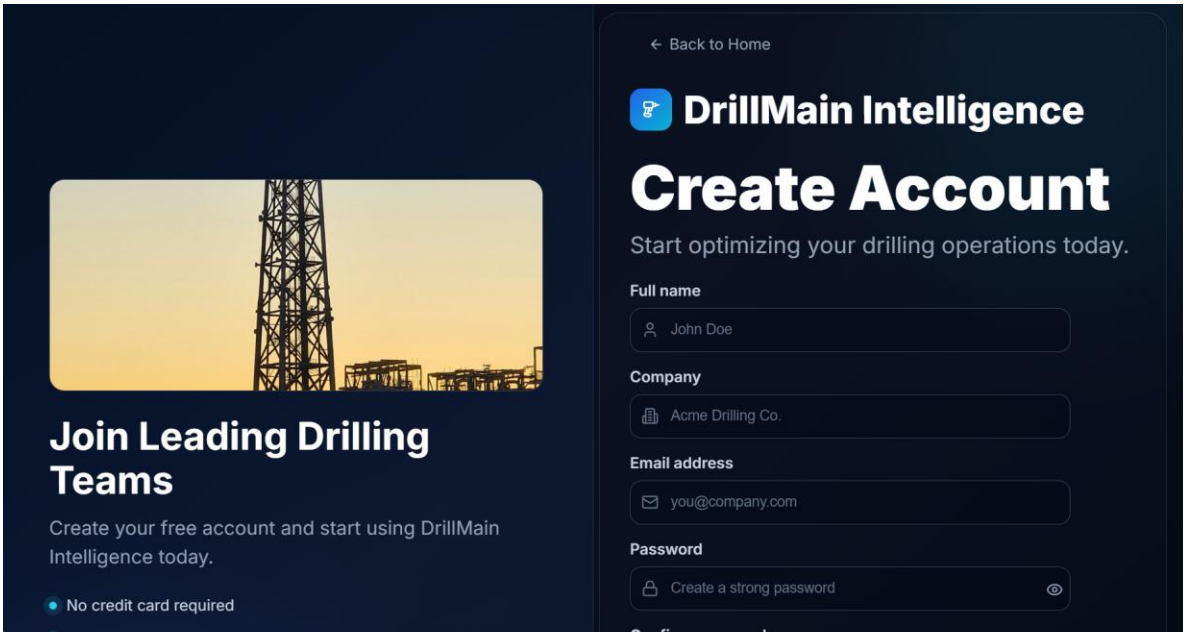
- On the Home page, click on “About” link at the top of the page. This page explains what DrillMain Intelligence gives you a brief introduction of what the app is all about.



Registration

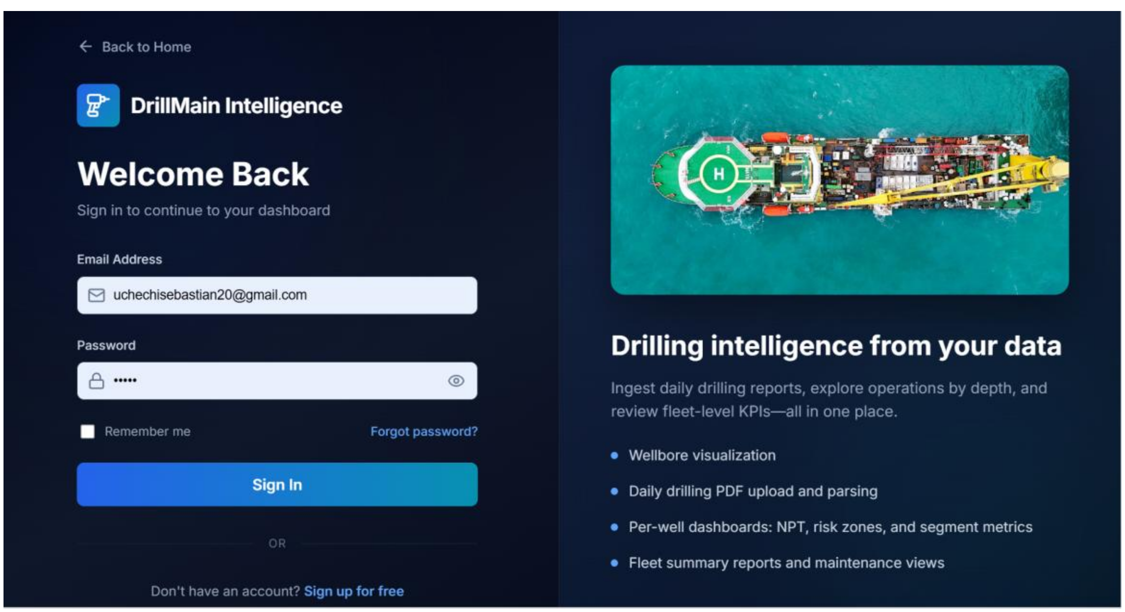
2.1 The sign-up page.

- On the home page, click on the sign-up button to create an account, by inputting your details.



2.2 The login page

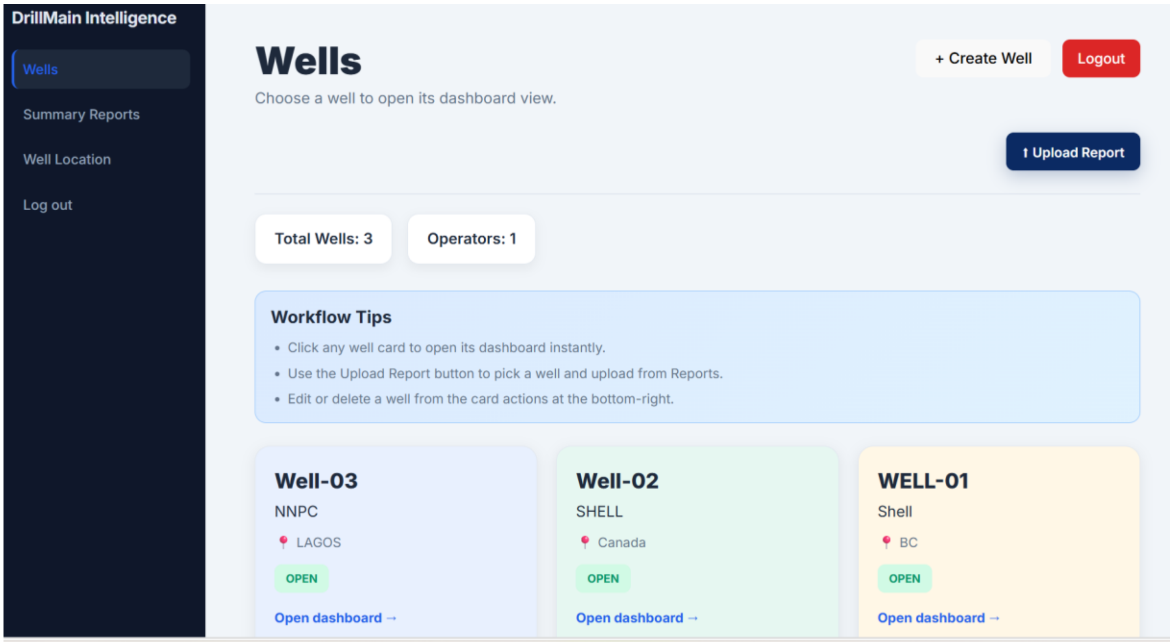
- After signing up, you will be directed to the login page.
- Fill in your login details to be able to login.



Creation of Wells

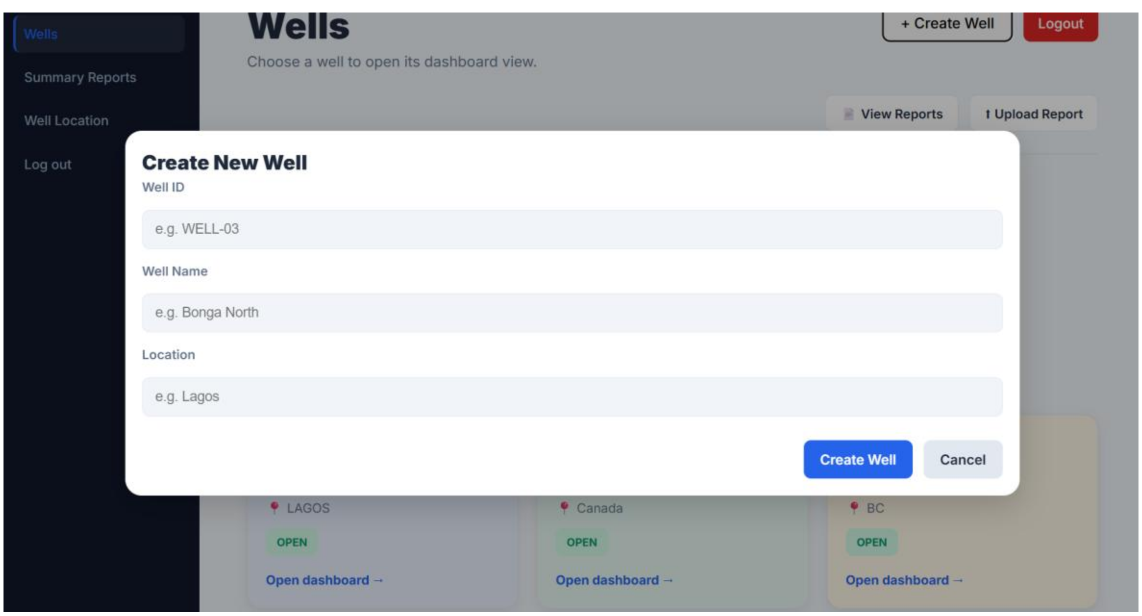
3.1 Wells Page

- After you login, you will be directed to the wells page.



3.2 Create New Well

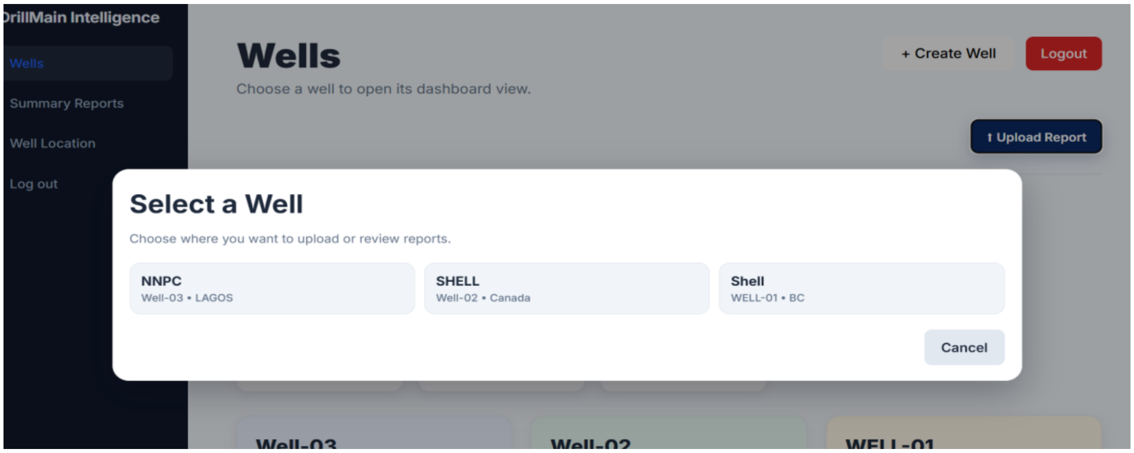
- Click on the create well button to register a new well.
- Fill the well ID, well name and location.
- Click "create well" button to create a new well.



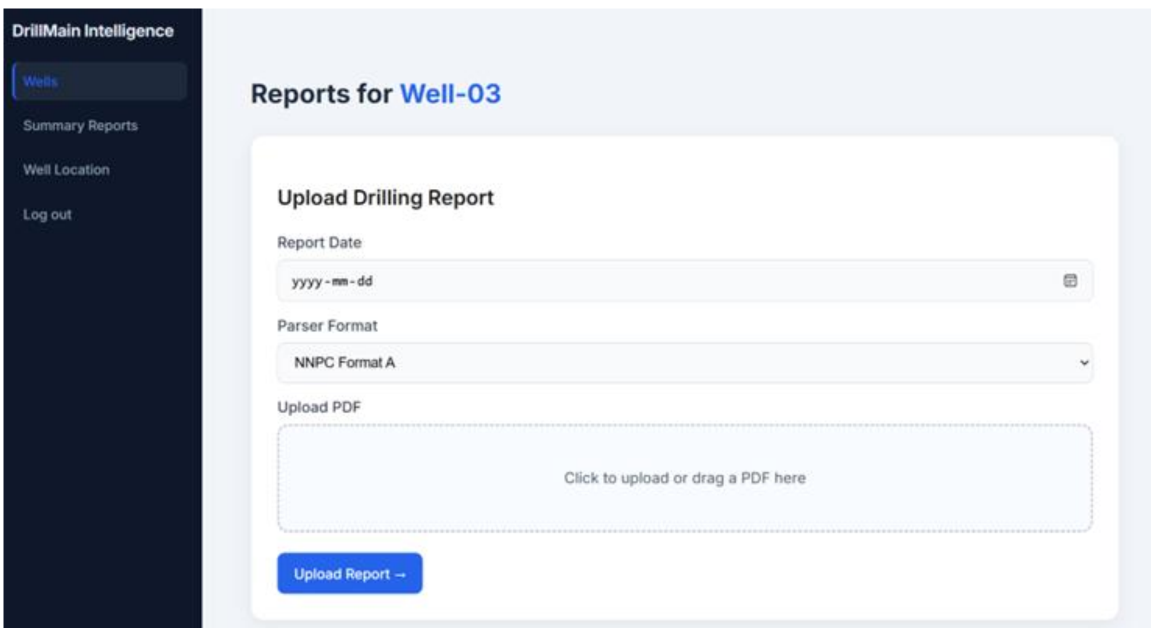
Reports

4.1 uploading Reports

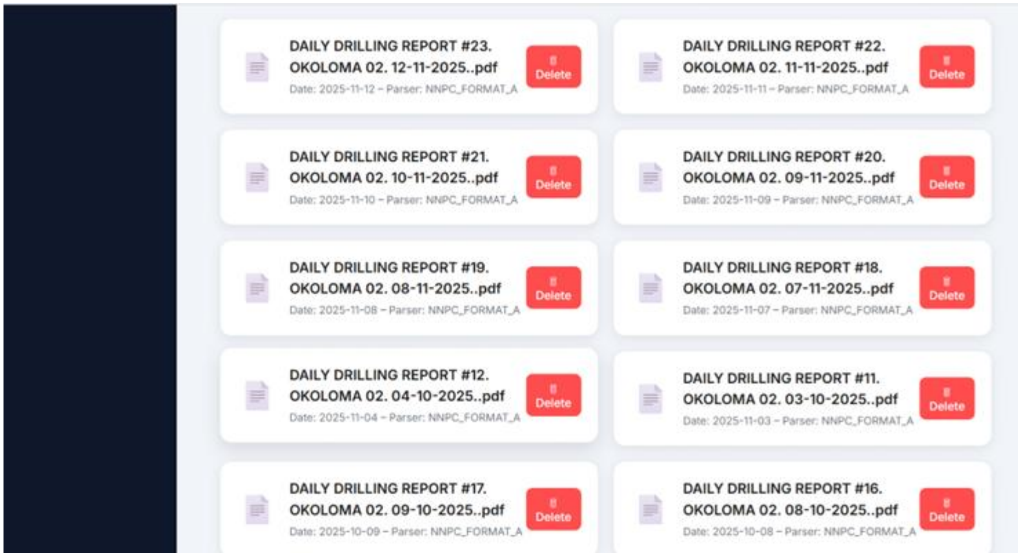
- On the wells page, click on upload report. This will open a window showing the list of wells, click on the well you would like to upload your report.



- After selecting a well, you will be directed to the reports page where you will upload your report.
- Fill in the report date choose the format and upload your pdf.



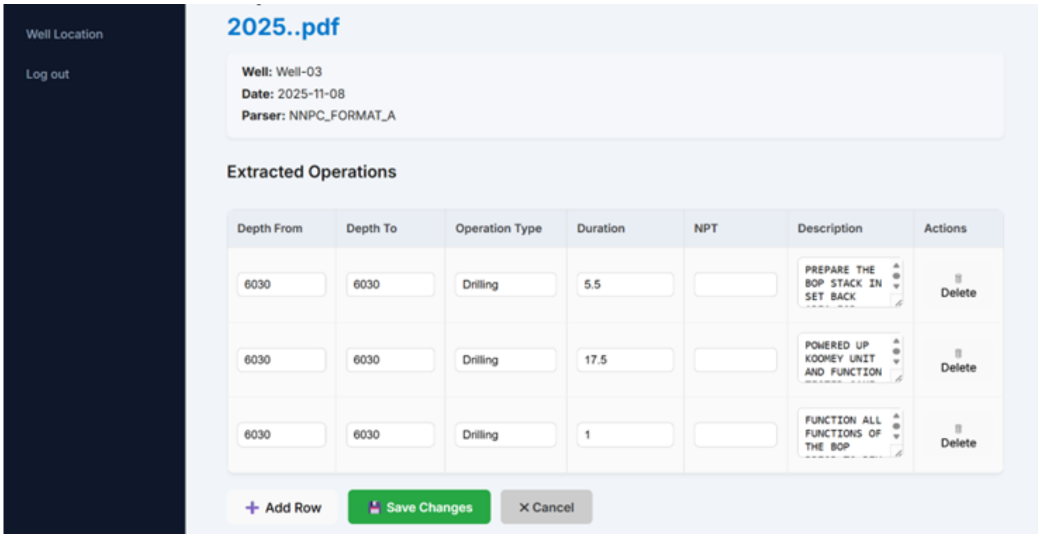
- Scroll down on the report page to see the uploaded reports.



4.2 Report Details / Editing report

- Click on the report to view the details of the report.

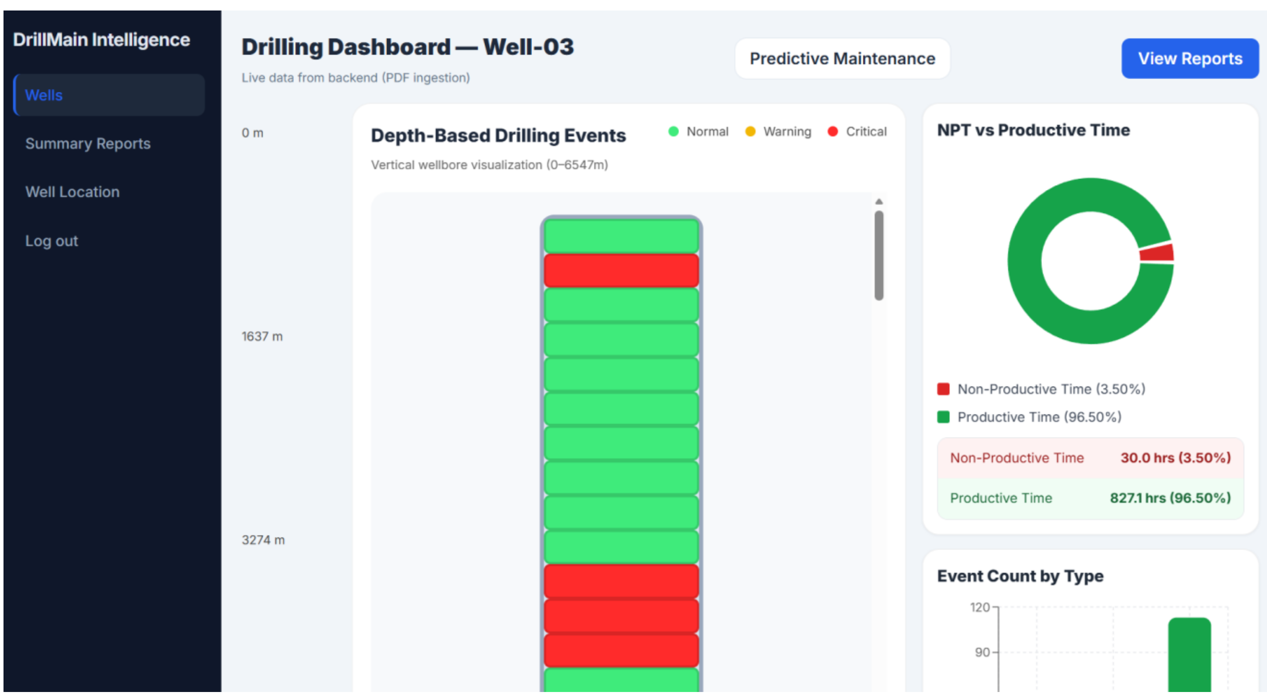
- Click on the “edit report” button to make changes if necessary.
- After clicking on the edit report button, the edit page will open to make changes on the report or delete data.
- Click on “save changes” after making changes to the report.

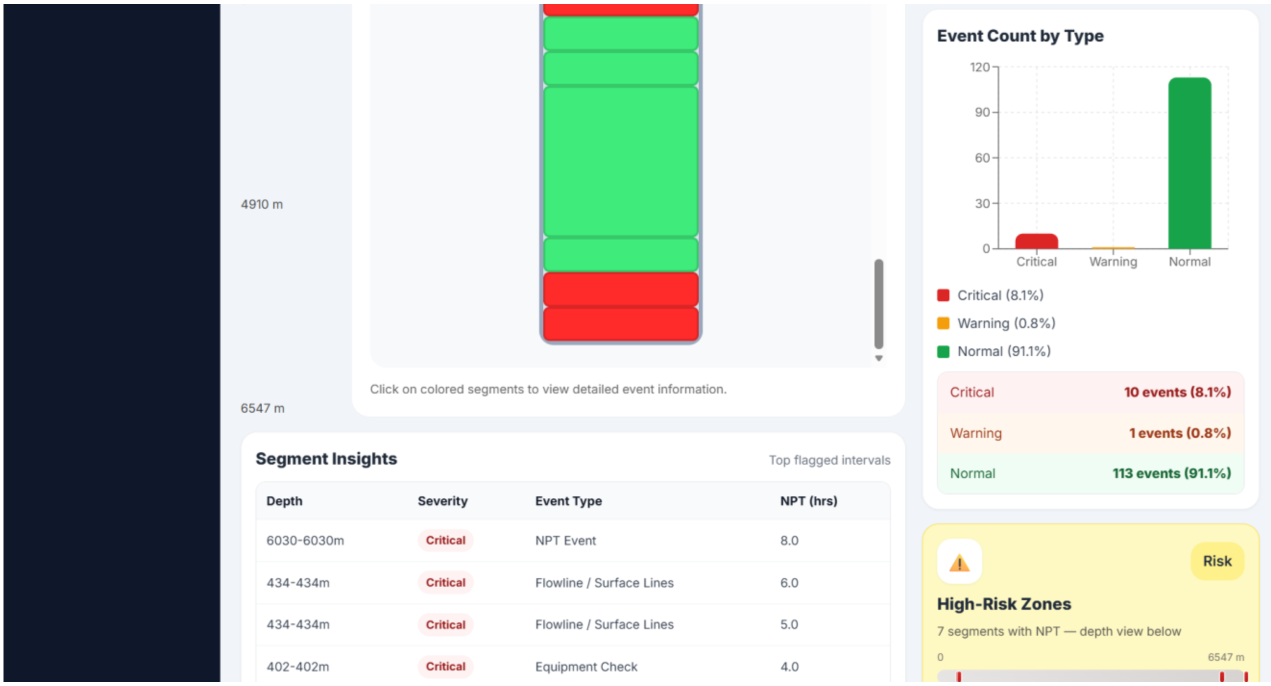


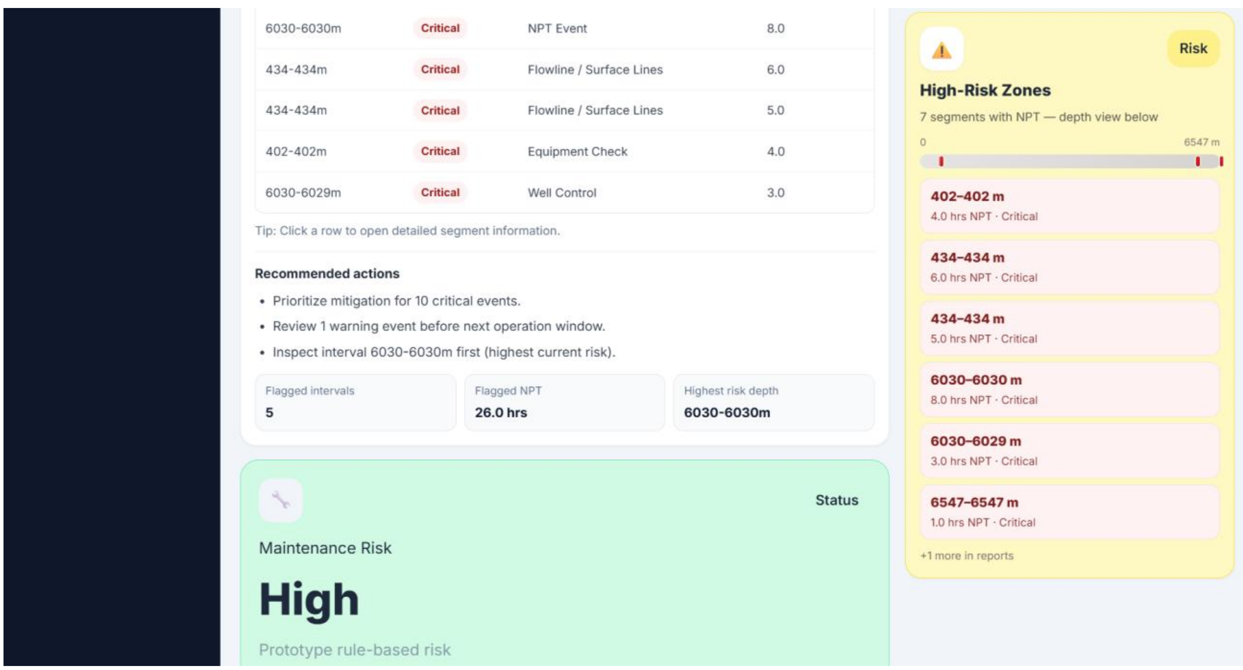
Dashboard

5.1 Drilling Dashboard

- After uploading and editing the report, go to the wells page and click on the well to view the dashboard.
- The dashboard shows a wellbore drill pipe visualization with segments (the blocks in the vertical pipe), which represents events that occurred in the uploaded report.

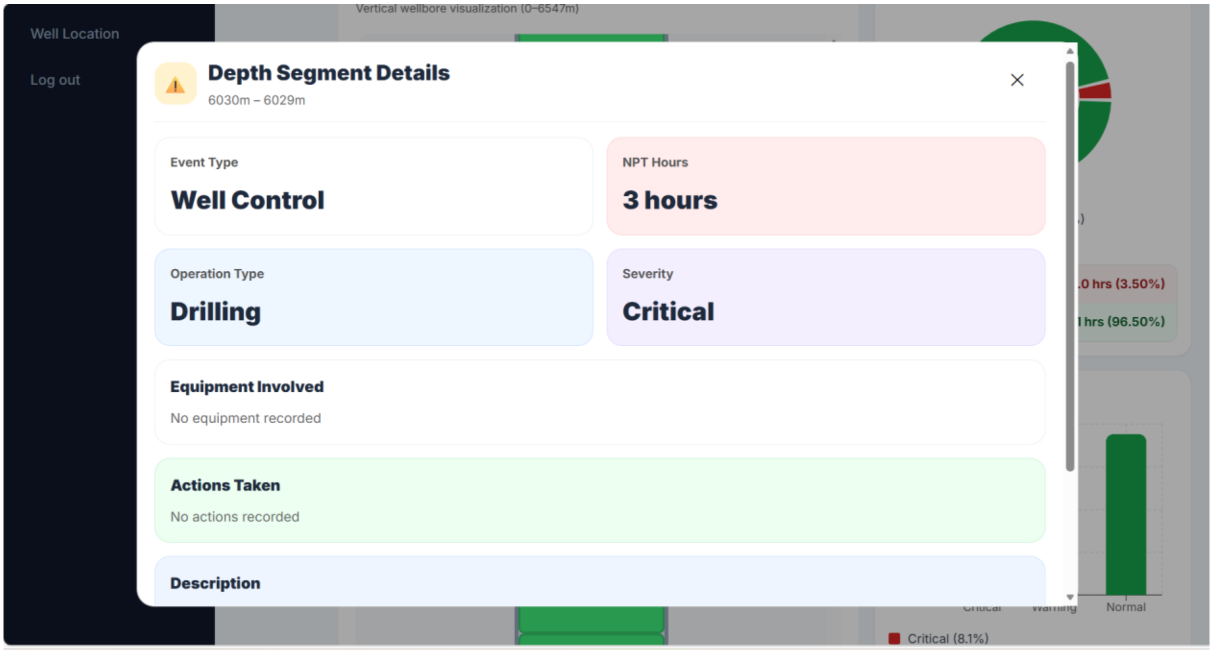



4910 m
6547 m
Click on colored segments to view detailed event information.
Segment Insights
Top flagged intervals
Depth
Severity
Event Type
NPT (hrs)
6030-6030m
Critical
NPT Event
8.0
434-434m
Critical
Flowline / Surface Lines
6.0
434-434m
Critical
Flowline / Surface Lines
5.0
402-402m
Critical
Equipment Check
4.0
Event Count by Type
120
90
60
30
0
Critical
Warning
Normal
Critical (8.1%)
Warning (0.8%)
Normal (91.1%)
Critical
10 events (8.1%)
Warning
1 events (0.8%)
Normal
113 events (91.1%)
Risk
High-Risk Zones
7 segments with NPT — depth view below
0
6547 m

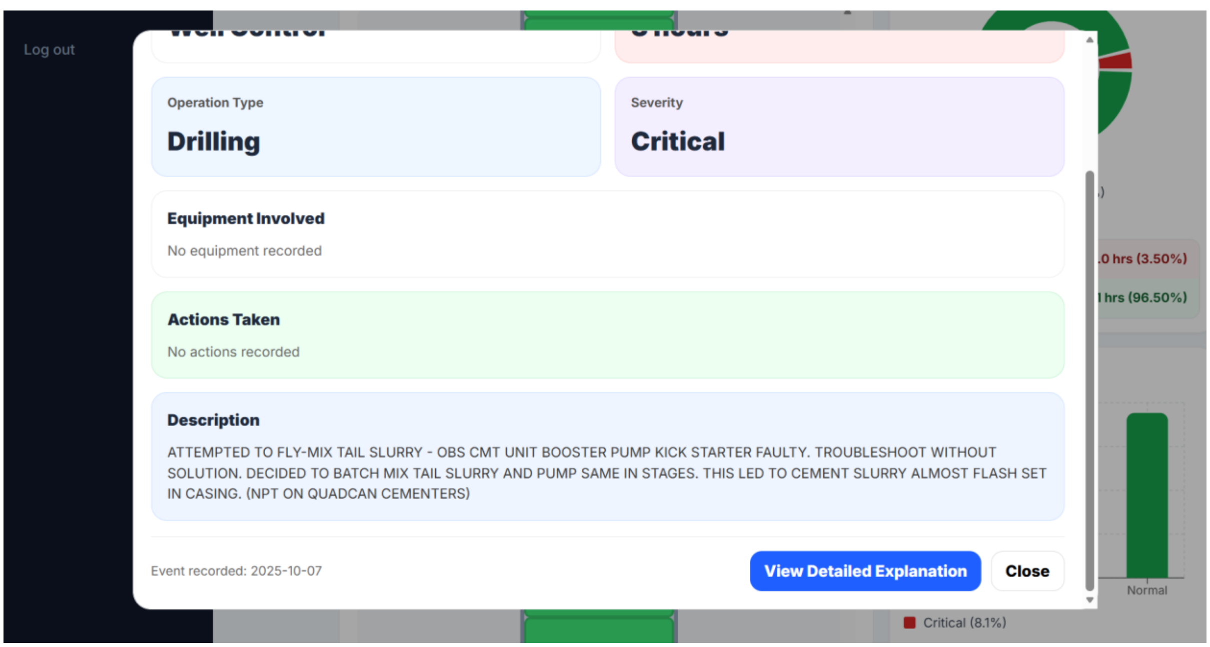

6030-6030m
Critical
NPT Event
8.0
434-434m
Critical
Flowline / Surface Lines
6.0
434-434m
Critical
Flowline / Surface Lines
5.0
402-402m
Critical
Equipment Check
4.0
6030-6029m
Critical
Well Control
3.0
Tip: Click a row to open detailed segment information.
Recommended actions
Prioritize mitigation for 10 critical events.
Review 1 warning event before next operation window.
Inspect interval 6030-6030m first (highest current risk).
Flagged intervals
5
Flagged NPT
26.0 hrs
Highest risk depth
6030-6030m
Status
Maintenance Risk
High
Prototype rule-based risk
Risk
High-Risk Zones
7 segments with NPT — depth view below
0
6547 m
402–402 m
4.0 hrs NPT · Critical
434–434 m
6.0 hrs NPT · Critical
434–434 m
5.0 hrs NPT · Critical
6030–6030 m
8.0 hrs NPT · Critical
6030–6029 m
3.0 hrs NPT · Critical
6547–6547 m
1.0 hrs NPT · Critical
+1 more in reports

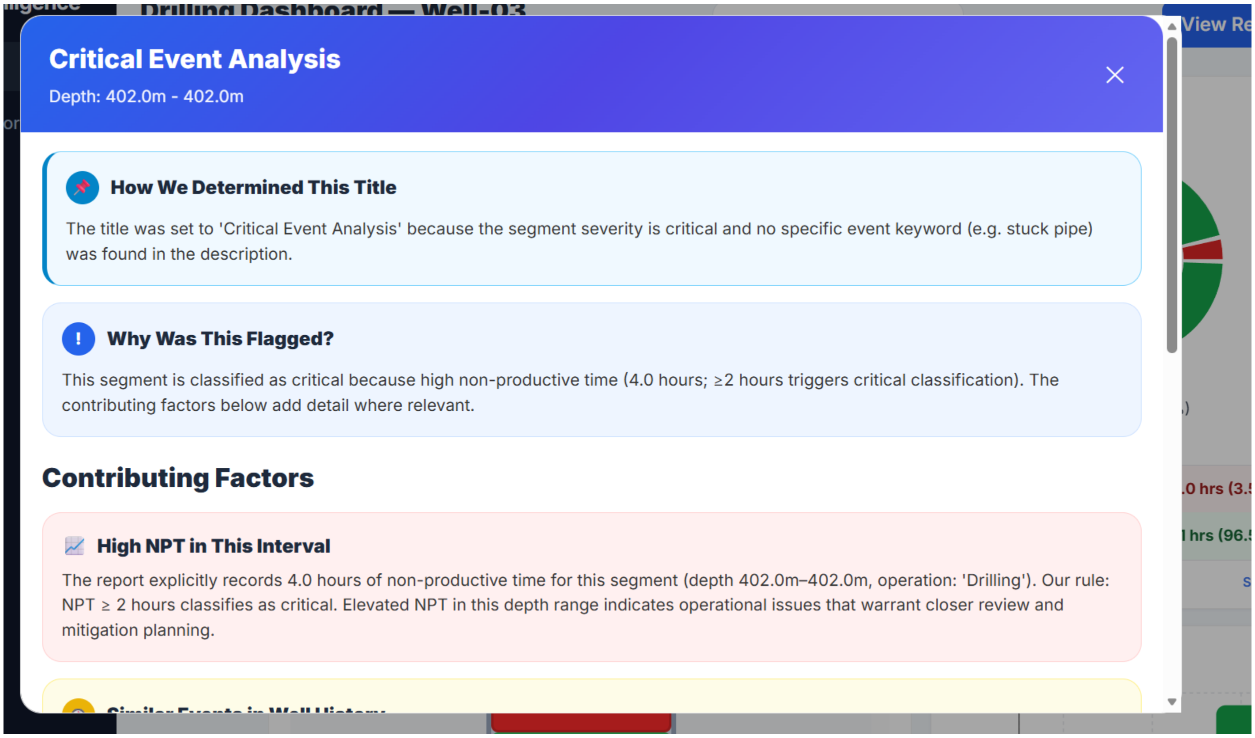
5.2 Segment Details Panel

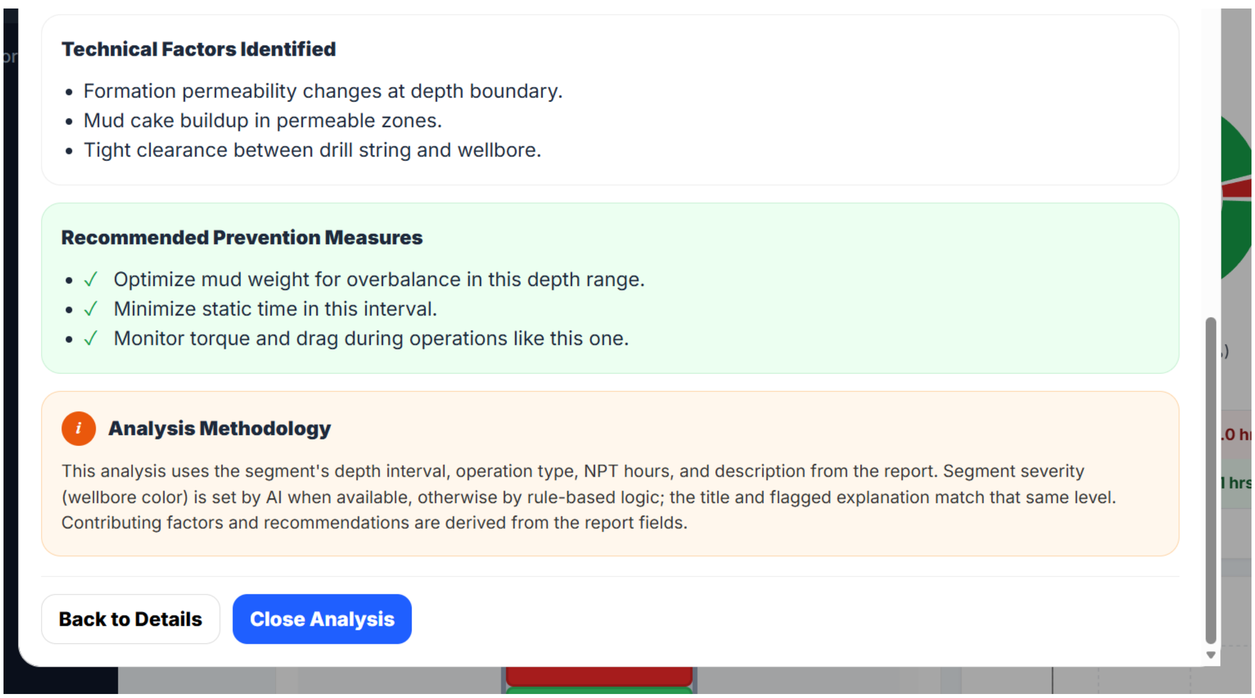
- Click on the segment and a segment modal will pop up showing a summary of what went on at a particular depth, the severity and the type of event that occurred.



- Click on "View Detailed Explanation" to view the analysis of the event that occurred at a particular depth.







- Click on “Back to details” to go back to the segment modal or click on “Close Analysis” to go back to the dashboard.

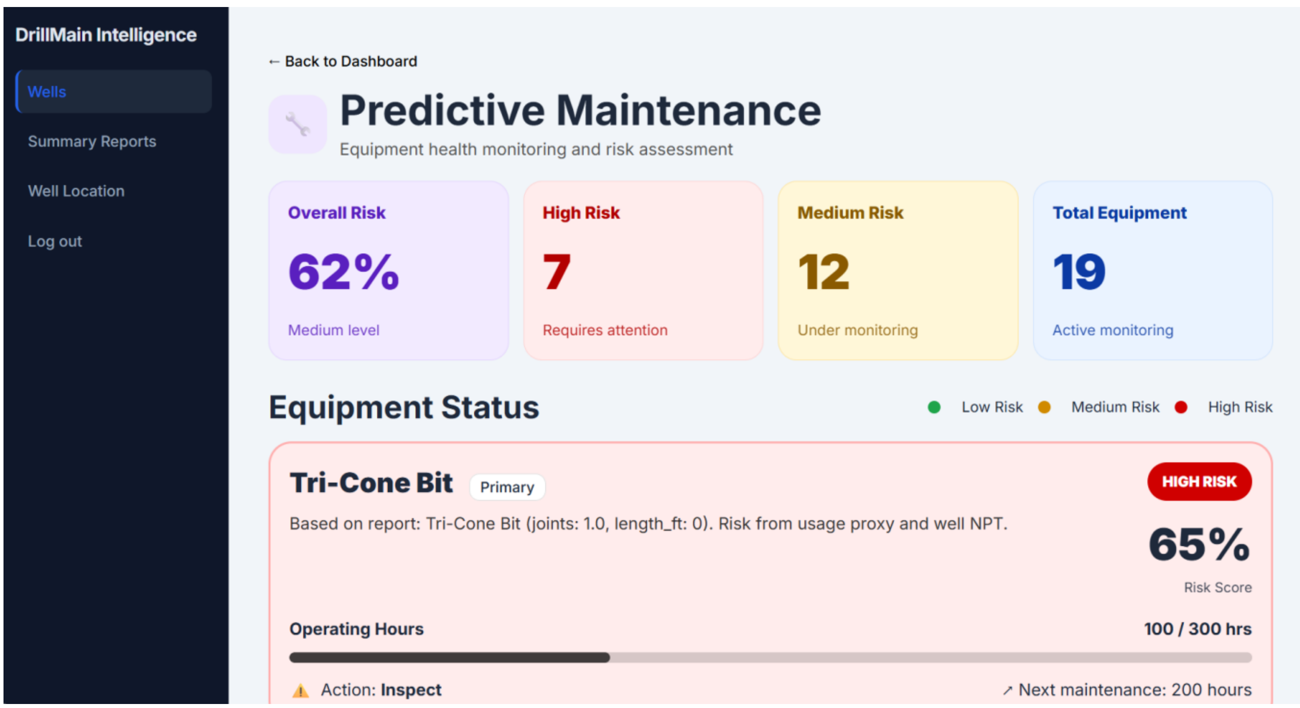
5.3 KPI Cards

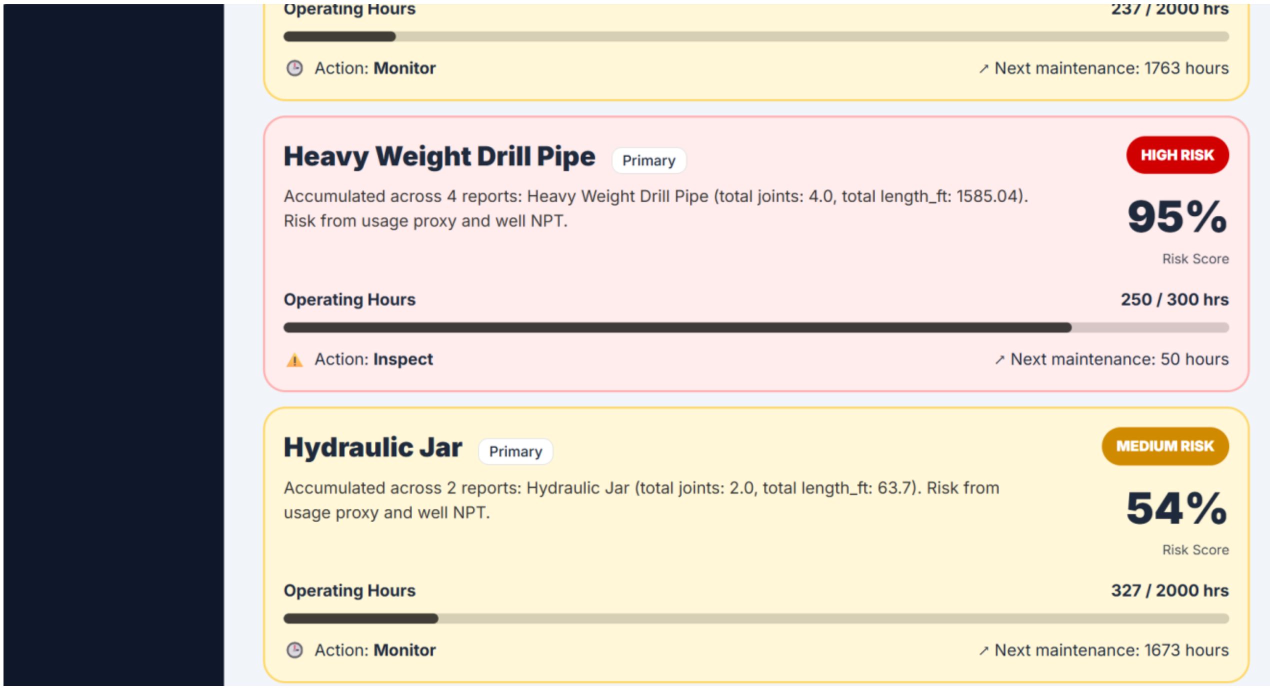
- Click on the **NPT vs Productive Time** KPI card on the dashboard to view more information on the Non-productive time.
- Click on the **Event Count by Type** KPI card on the dashboard to view more information on the event count based on the type of event that occurred.

Prédictive Maintenance

6.1 Prédictive Maintenance Page

- Click the predictive maintenance button in dashboard page to open maintenance page for your equipment.



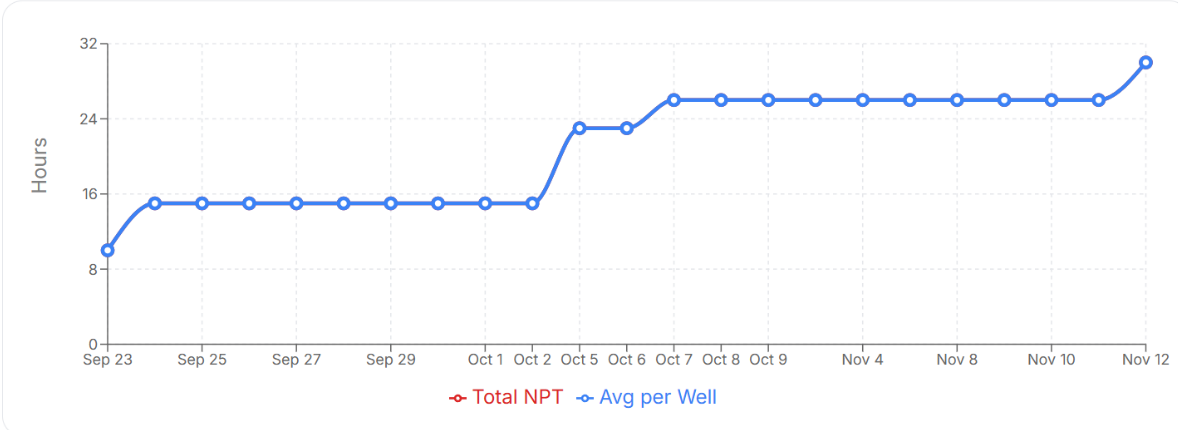


Summary Reports

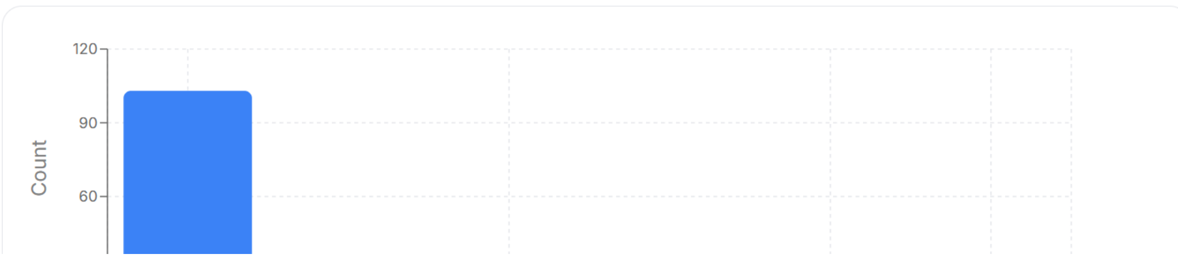
6.1 Summary Reports page

- Click on “Summary Reports” on the dashboard page in the layout section, to see the summary details of all wells in your account.

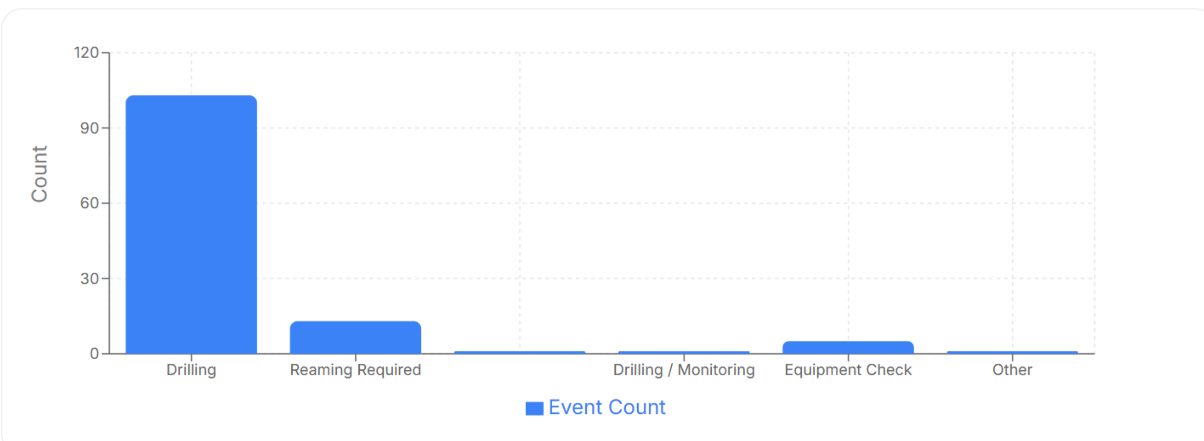
Fleet NPT Trend (Cumulative)



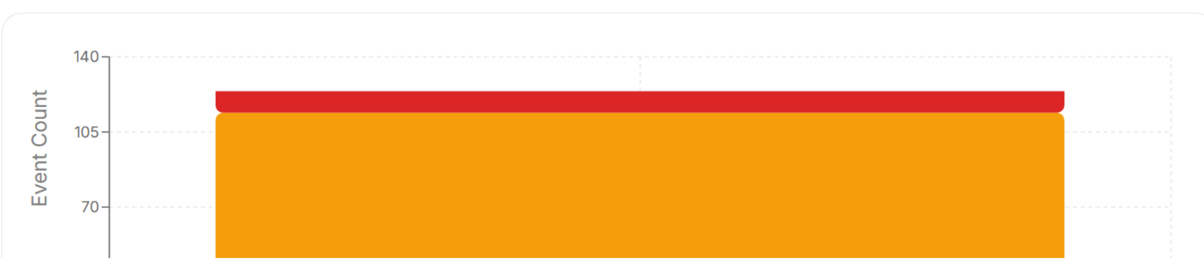
Event Type Distribution (All Wells)



Event Type Distribution (All Wells)



Events per Well



Location

7.1 Well Location page

- Click on “Well Location” on the dashboard page in the layout section, to see the location of all wells in your account on a map.